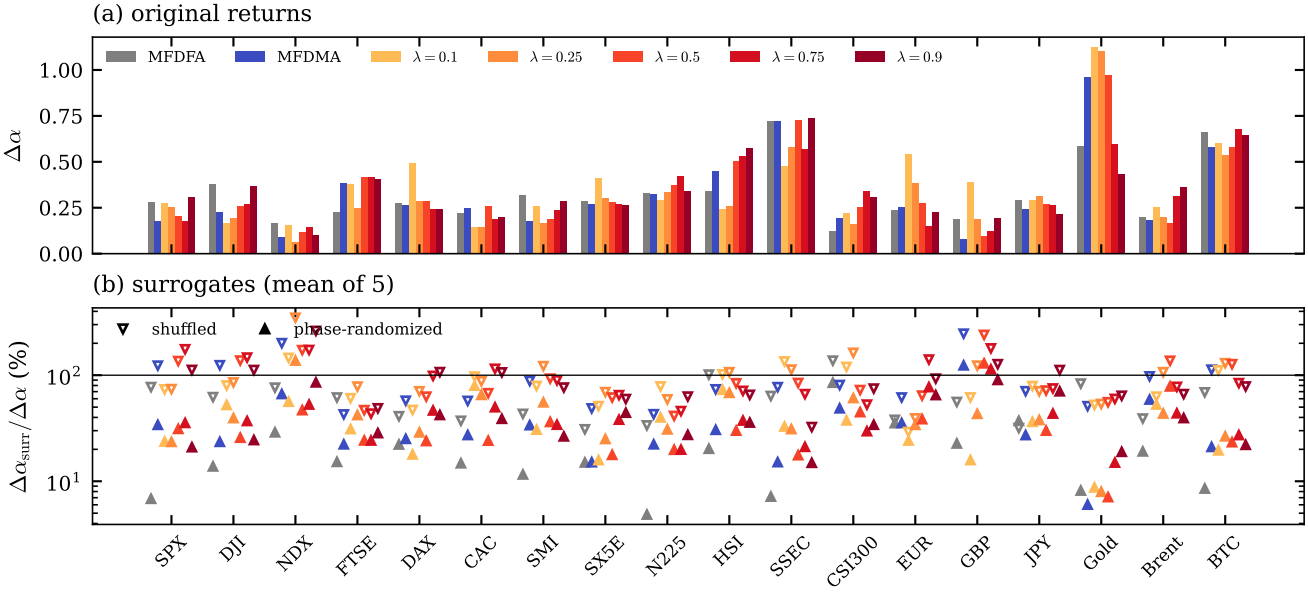
(a) original returns
MFDFA
MFDMA
$\lambda = 0.1$
$\lambda = 0.25$
$\lambda = 0.5$
$\lambda = 0.75$
$\lambda = 0.9$
$\Delta\alpha$
1.00
0.75
0.50
0.25
0.00
(b) surrogates (mean of 5)
shuffled
phase-randomized
$\Delta\alpha_{\mathrm{surr}}/\Delta\alpha$ (%)
10^2
10^1
SPX
DJI
NDX
FTSE
DAX
CAC
SMI
SX5E
N225
HSI
SSEC
CSI300
EUR
GBP
JPY
Gold
Brent
BTC